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# Feature Quality Assessment: A Database and A Lightweight Objective Method

Shipei Wang, Ping An, *Member, IEEE*, Chao Yang, *Member, IEEE*, Gongyang Li, *Member, IEEE*, Xinpeng Huang, *Member, IEEE*, and Shiqi Wang, *Senior Member, IEEE*

***Abstract*—In the era of Artificial Intelligence, visual data gathered by edge devices could be primarily utilized for machine vision tasks. The prominent coding frameworks accomplish this by extracting and compressing features extracted from input data. As such, the quality of these features is vital, as they reflect the performance of the coding framework. However, much less work has been dedicated to quality assessment on features, impeding the optimization of the coding system. In this work, we pioneer to explore the feature quality assessment by creating a novel database tailored for features, with the quality ground-truth for each feature. Then, we propose a lightweight feature quality assessment method, called Lightweight Feature Quality Assessment (LFQA). We analyze the feature characteristics from the perspective of spatial and channel thoroughly, and the framework of LFQA is designed based on the analysis results. Experimental results demonstrate that LFQA accurately evaluates the quality of features, reaching a notable Spearman Rank-Order Correlation Coefficient of 85.38%, and exhibits competitive performance in improving the performance of video coding for machine system. Furthermore, LFQA has fewer model parameters and faster inference speed, ensuring a wide range of promising applications.**

***Index Terms*—Video coding for machine, feature quality assessment, lightweight framework, feature characteristics.**

## I. Introduction

ADVANCEMENTS in artificial intelligence (AI) are accelerating the development of intelligent applications, such as image classification and object detection. Therefore, the ultimate consumers of visual data are experiencing a shift from human vision to machine vision. Traditional coding frameworks, such as High Efficiency Video Coding (HEVC) [1] and Versatile Video Coding (VVC) [2], are designed for human vision, which differs significantly from machine vision. Therefore, new coding frameworks need to be developed to satisfy the requirements of both human vision and machine vision simultaneously.

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S. Wang, P. An, C. Yang, G. Li, X. Huang are with the Key Laboratory of Specialty Fiber Optics and Optical Access Networks, Joint International Research Laboratory of Specialty Fiber Optics and Advanced Communication, Shanghai Institute for Advanced Communication and Data Science, and School of Communication and Information Engineering, Shanghai University, Shanghai 200444, China (email: wang_shipei0824@shu.edu.cn; anping@shu.edu.cn; yangchaoie@shu.edu.cn; ligongyang@shu.edu.cn; huangxinpeng@shu.edu.cn).

S. Wang is with the Department of Computer Science, City University of Hong Kong, Kowloon, Hong Kong (email: shiqwang@cityu.edu.hk).

Corresponding author: Ping An

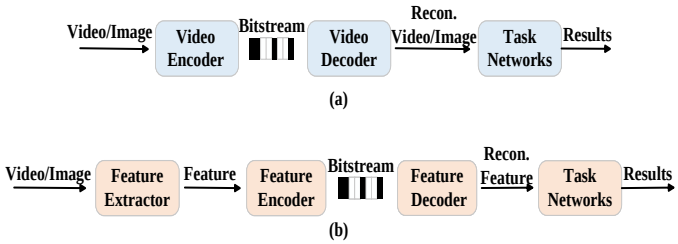


Fig. 1. Two pipelines of VCM. (a) Compression then analysis; (b) Analysis then compression.

Relevant standardization activities, such as Video Coding for Machine (VCM) [3] and JPEG-AI [4], have been initiated to tackle the aforementioned challenge. VCM encompasses two primary pipelines, including compression then analysis (CTA) and analysis then compression (ATC), as depicted in Fig. 1. In CTA, images undergo compression and are transmitted to the decoder, where tasks are implemented on the decoded images. In ATC, tasks are completed in the feature domain, where features are initially extracted from the video/image signal, subsequently compressed, and ultimately transmitted to the decoder for further processing. This pipeline facilitates the transfer of partial workload from the decoder to the encoder, termed Collaborative Intelligence (CI) [5], further optimizing system latency and improving resource utilization efficiency against CTA. Consequently, ATC has gradually emerged as the focal point of research.

In ATC, as features are forwarded to the decoder for various tasks, they directly affect task performance and become one of the key points for the coding system. In order to alleviate the transmission burden, these features undergo compression before being transmitted to the decoder, resulting in a decline in task performance. It is worth noting that adversarial examples can also affect features and subsequently impact task performance. However, in VCM, the primary focus is on balancing compression distortion and task performance [6]. Therefore, to advance the development of VCM, our work primarily considers the impact of compression distortion on features. Fig. 2 illustrates the rate-distortion performance and rate-accuracy performance on human and machine tasks, as reported in [7]. In this study, image reconstruction is designated as the human task, while image classification is designated as the machine task. Notably, as the bits per pixel (BPP) diminishes, both human and machine task performances exhibit a corresponding decrease. Therefore, accurately quantifying the distortion of

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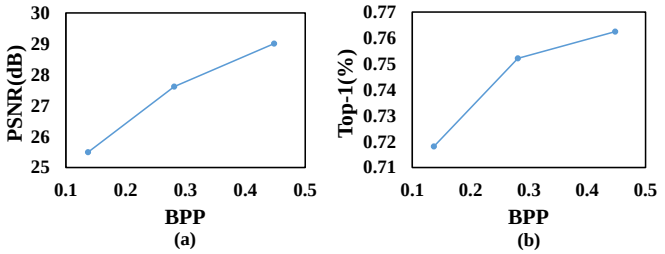


Fig. 2. The performances of human task and machine task. (a) The rate-distortion performance of image reconstruction; (b) The rate-accuracy performance of image classification.

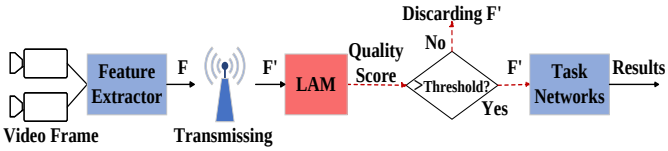


Fig. 3. Illustration of the application of LAM. More specifically, F′ is fed into the LAM module, from which the quality score is derived. In practical application scenarios, an acceptable performance threshold is established for the coding system. If the quality score does not reach the threshold, this feature is discarded. The filtered F′ is passed to the subsequent modules, thereby improving the coding system performance.

features caused by compression contributes to distinguish which features are beneficial to the ATC coding system and plays an important role in optimizing the performance of the coding system.

Hence, we present the concept of feature quality, which reflects the distortion degree of features, where the feature quality negatively correlates with the distortion degree of features. A lightweight and accuracy assessment method (LAM) for feature quality can serve as a plug-and-play module that can be flexibly embedded into the ATC coding systems, optimizing the coding performance. Fig. 3 shows one specific case. Inspired by [8], LAM is regarded as a filtering module and integrated into the ATC coding system. LAM can filter out features with poor quality, thereby improving the ATC coding system performance. The pipeline of this case is suitable for many security scenarios, such as facial recognition and vehicle detection, demonstrating the significant potential of LAM.

In summary, our goal is to systematically assess the feature quality. Image quality assessment (IQA) has been extensively developed over several years, yielding numerous classic and efficient IQA methods. However, features represent a different data format with distinct statistical characteristics compared to images [9]. Directly applying existing IQA methods to assess feature quality may not yield optimal outcomes. Furthermore, feature quality assessment (FQA) is in the preliminary exploration stage. To the best of our knowledge, [10] is the only published FQA work (CFQA). Specifically, CFQA develops distinct FQA methods for three different machine vision tasks (e.g., classification, retrieval, and object detection). However, there are scenarios in VCM that require human observation. To address this issue, we need to consider both machine vision tasks and image reconstruction when developing the FQA methods. Hence, there is an urgent need for novel feature quality assessment methods tailored specifically for humans and machines. Inspired by the existing IQA methods, we first construct a database for features, then assign the Ground-Truth, which is similar to the human opinion score in IQA, to each feature through the indicator proposed for humans and machines. Finally, we propose the LFQA, a lightweight feature quality assessment method for human and machine tasks. In LFQA, the characteristics of features are analyzed, which serves as the main motivation behind the proposed approach. We introduce a Pre-processing module and a Channel Selector module based on the analysis results.

In this paper, we make the following contributions:

- We conduct a preliminary exploration on feature quality assessment and construct the first feature database (FDB).
- We present a new indicator $S_{CI}$ to assess the feature quality by considering both the human and machine task performance. The Ground-Truths of the FDB are obtained through the indicator. $S_{CI}$ is one of the few indicators that simultaneously consider the humans and machines, fostering the progression of VCM.
- We design a lightweight deep-learning-based feature quality assessment framework named LFQA and consider the characteristics of features when designing the framework by introducing a Pre-processing module and a Channel Selector module. LFQA runs fast and has few module parameters and low computational complexity, making it highly suitable for real-time applications.

The rest of this paper is organized as follows. In Sec. II, we review the related work of video coding for machine and image quality assessment. In Sec. III, we introduce our feature database. In Sec. IV, we elaborate our method in detail. In Sec. V, we present adequate experiments and analysis. In Sec. VI, we summarize the conclusion.

## II. Related Work

In this section, we review classical works of video coding for machine and image quality assessment.

### A. Video Coding for Machine

Owing to the advanced 5G and AI technologies, an increasing amount of video/image data captured by edge devices like surveillance cameras and smart phones is consumed by machines. Given the immense volume of the data, effective compression methods are essential to reduce the transmission load. Existing compression methods, such as JPEG [11], HEVC, and VVC, are designed for the Human Visual System (HVS) and aim to achieve better rate-distortion performance between compression ratio and image quality. However, the objectives and performance metrics of machine vision tasks differ from those of human vision tasks, necessitating the proposal of new coding frameworks for machine vision. The aforementioned VCM aims to carry out a series of standardization activities for machine vision.

In Fig. 1 (a), traditional coding algorithms such as HEVC, VVC, or end-to-end compression methods [12]–[15] are utilized to encode input videos/images, with subsequent machine vision tasks performed on the decoded videos/images. [16]

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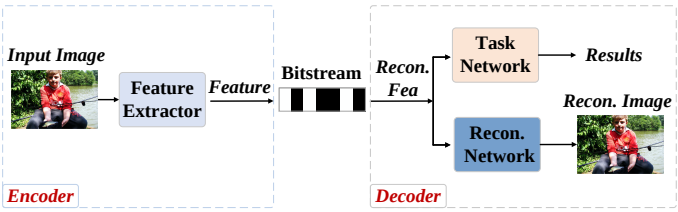


Fig. 4. The framework of CDF. *Input Image* is fed into the Feature Extractor to obtain the *Feature*. *Feature* is compressed and transmitted to the *Decoder*, and the *Recon.Fea* is got. Feeding the *Recon.Fea* to the Task Network and Recon.Network at the same time, the results of human and machine tasks are gained.

proposes a semantic metric and integrates it into an end-to-end image compression framework, optimizing the performance for both human and machine tasks. [17] proposes a lightweight image coding framework for humans and machines. Specifically, this work introduces a spatial-frequency modulation adapter module, which enhances the rate-distortion performance in both spatial and frequency domain. [18] is the first attempt to embed prompting into the image coding frameworks, enabling the transfer of human perception based coding frameworks to machine perception without the need for fine-tuning the frameworks.

In Fig. 1 (b), features extracted from input videos/images are encoded by traditional coding algorithms or end-to-end compression methods. On the decoder side, the reconstructed features are directly fed into subsequent task networks to complete the required tasks. Recently, more and more research has focused on the pipeline depicted in Fig. 1 (b), leading to the development of coding frameworks tailored for VCM. For instance, [19] proposes a scalable image coding framework for face images, where the base layer is optimized for machine tasks, while the enhancement layer facilitates high-quality image reconstruction. [20] embeds an advanced generative model into a scalable compression framework to achieve better rate-distortion performance. [21] analyzes feature redundancy from spatial and statistical perspectives and proposes a low-rate feature compression method. [22] introduces the concept of "feature scalable", where features are partitioned into several parts along the channel dimension, with partial features allocated for machine vision tasks and entire features utilized for human vision tasks. [7] integrates the Transformer [23] into the coding framework, benefiting both human and machine tasks. Other work [24] introduces the prompt module originating from NLP (Natural Language Processing) into the coding framework, improving task performance through training the prompt module in an end-to-end manner. Features are a critical component of the ATC framework, and their quality determines the ATC performance. Therefore, accurately assessing feature quality is essential for enhancing the performance of ATC and advancing the development of VCM.

### B. Image Quality Assessment

Over the years, many studies have focused on image quality assessment. IQA aims to quantify perceptible image distortion by the human eye. Extracting effective features is crucial for IQA. Early traditional works focus on extracting manual features from images, including SSIM [25], MS-SSIM [26], FSIM [27], PSNR [28], and VIF [29]. With the development of deep learning methods, researchers have begun utilizing deep neural networks to extract deep features. Classical machine vision networks such as Resnet50 [30], VGG [31], MobileNet [32], and EfficientNet [33] are commonly employed as feature extractors. Compared with the manual features, deep features have demonstrated better performance. Hence, deep-learning-based IQA methods have gradually replaced traditional IQA methods as the predominant approach. There are many deep-learning-based IQA methods such as NADN [34], SFA [35], GraphIQA [36], TReS [37], MUSIQ [38], CVC [39], CKDN [40], USTS [41], and LPIPS [42]. Recently, studies have introduced the multi-modality large language models (MLLMs) into IQA, [43] trains the MLLMs to perform visual rating in alignment with human perception. [44] introduces the Q-Boost strategy to enhance low-level MLLMs in IQA. These methods inject new vitality into the field of IQA.

Besides IQA methods, universal and efficient databases are also important for IQA, as they can effectively reflect the performance of IQA methods, thereby improving them. In addition, they also contribute to training more versatile IQA methods. Several classical IQA databases have been built, such as TID2013 [45] and LIVE [46]. Recently, several databases targeting MLLMs have emerged [47], [48]. These databases contain images with various types, which are useful for increasing the robustness of IQA methods.

However, the IQA methods and databases mentioned above are all designed for human perception while ignoring semantic information critical for machine vision. Furthermore, the inputs of IQA methods are images with three channels, whereas features extracted from deep neural networks typically have more than three channels. In summary, directly applying existing IQA methods to evaluate the quality of features is difficult, and the evaluation results may not be satisfactory. Therefore, it is necessary and urgent to develop databases specifically oriented toward features and to propose FQA methods that evaluate both human and machine task performances based on these feature databases. Hence, we firstly build a feature database and present an indicator to measure the performance of human and machine tasks. The Ground-Truths of our feature database are obtained through this indicator. Then we thoroughly study the spatial and channel statistical characteristics of features and finally design a lightweight feature quality assessment method based on these findings. This method accurately evaluates the quality of features and is flexibly embedded into the coding system to optimize it.

## III. Feature Quality Assessment Database

In this section, we present our feature database in detail. In Sec. III-A, we introduce the process of obtaining feature data. In Sec. III-B, we propose an indicator considering both human and machine tasks simultaneously.

VCM is currently in a rapid development stage, with various coding frameworks emerging, leading to significant differences in feature statistical characteristics across these frameworks.

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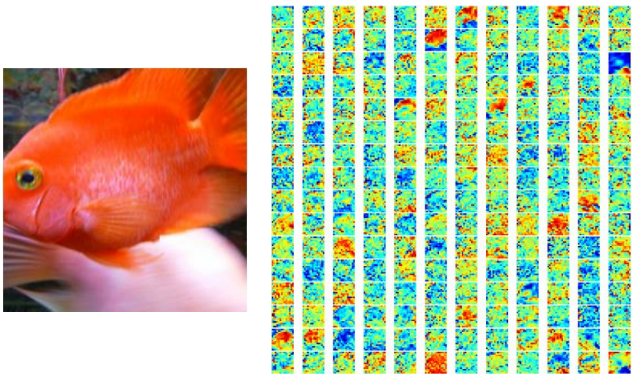

(a) (b)

Fig. 5. The visualization result of one feature. (a) The original image from the ImageNet database; (b) The visualization of the feature corresponding to (a). Each small square represents a feature channel, totaling 192 channels.

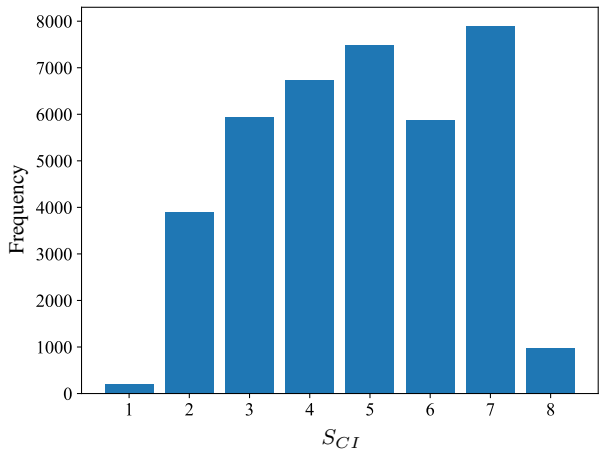


Fig. 6. Illustration of the histogram which shows the distribution of $S_{CI}$ values across the FDB.

However, as an early exploratory work on FQA, we focus more on validating the feasibility and necessity of FQA. Therefore, in this work, we concentrate on the classic ATC coding framework (CDF) [7], using it as the foundation to establish our feature database and explore the relationship between feature quality and task performance.

### A. Feature Extraction

As mentioned above, we choose CDF, as illustrated in Fig. 4, to obtain the feature data and the corresponding Ground-Truths. We execute the CDF with pre-trained weights, while maintaining all parameters fixed, thereby generating and storing the *Recon.Fea*, which can support human and machine tasks simultaneously, at the *Decoder*. The CDF with different pre-trained weights exhibits varied rate-distortion performances, consequently yielding features with different compression ratios. In our database, to maintain consistency, the compression ratios of our features are aligned with those of the CDF, which are 0.137 BPP, 0.281 BPP, and 0.448 BPP respectively.

We select images from the validation set of the ImageNet database [49] as inputs for the CDF. Following the steps outlined in the previous paragraph, we obtain features corresponding to each image, resulting in 39000 features with three compression ratios in the feature database. Fig. 5 illustrates the visual result of the feature in FDB.

### B. A Quality Indicator for Features

A new indicator $S_{CI}$ is proposed to assess feature quality, which reflects the performance for both human and machine tasks. Additionally, the performance of the ATC coding system can also be evaluated through $S_{CI}$. Unfortunately, assessing feature quality directly is challenging, and as shown in Fig. 2, where increasing compression ratio of features degrades the rate-distortion performance of the ATC system. Therefore, we capture feature compression distortion by assessing the performance of the ATC system. In this way, we can indirectly assess the feature quality. Specifically, after acquiring the performance indicators for each task, $S_{CI}$ is obtained by integrating these individual task performance indicators through the following strategy:

$$S_{CI} = \lambda_1 * S_H + \lambda_2 * S_M, \tag{1}$$

where $S_H$ denotes the performance indicator for human task, and $S_M$ denotes the performance indicator for machine task. The parameters $\lambda_1$ and $\lambda_2$ represent the weights assigned to the two tasks.

In our work, we adopt the same tasks as those utilized in the CDF. For the human task, we use PSNR, a simple and efficient indicator to measure the quality of reconstructed images. Regarding the machine task, we utilize the Top-1 accuracy to measure the performance of image classification. In our work, the Top-1 accuracy is calculated for each feature, resulting in either 0 or 1. 0 represents failed classification, while 1 represents successful classification. It's important to note that PSNR and Top-1 accuracy have distinct meanings and differing value distributions. To appropriately weigh these two indicators, we normalize them to the range of 0-1 separately and fuse these two normalized values according to Eqn. (1) to derive the $S_{CI}$ for each feature. As an initial exploration, we assume that $\lambda_1$ equals $\lambda_2$ here. The $S_{CI}$ for each feature is represented as a decimal number with four decimal places. To facilitate a clearer interpretation of ATC coding system performance, each $S_{CI}$ is categorized into an integer from 1 to 8, with the specific range determined based on $S_{CI}$ values. Fig. 6 shows the distribution of $S_{CI}$, where extreme scores of 1 and 8 are relatively rare, while moderate scores are relatively common. The $S_{CI}$ distribution is similar to that of challenging IQA database [46].

## IV. Proposed Method

In this section, we present the LFQA in details. In Sec. IV-A, we introduce the network overview of the LFQA. In Sec. IV-B, we elaborate the proposed Pre-processing module. In Sec. IV-C, we elaborate the proposed Channel Selector module. In Sec. IV-D, we introduce the proposed Baseline Network.

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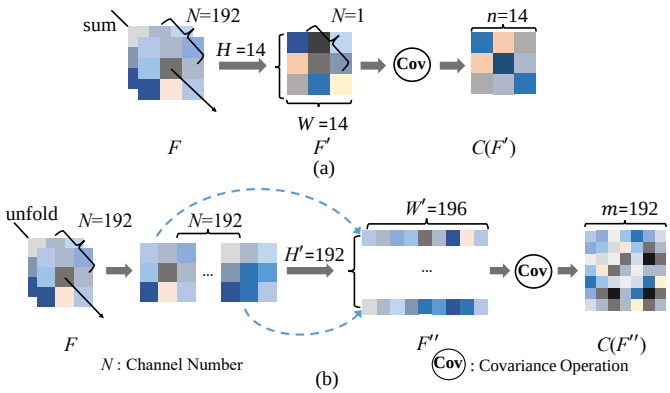


Fig. 7. The process of obtaining $C(F')$ and $C(F'')$. (a) Obtaining the $C(F')$; (b) Obtaining the $C(F'')$. Zoom-in for viewing details.

### A. Network Overview

The overall framework, as depicted in Fig. 8 (a), comprises three distinct modules. The Pre-processing module consists of a Spatial Attention module and a Channel Attention module. The three-dimensional input $F \in \mathbb{R}^{H\times W\times C}$, where $H$, $W$, and $C$ represent the input height, width, and the number of input channels respectively, is weighted by the Pre-processing module while preserving its original shape, yielding the $F_1$. To further eliminate redundant information, $F_1$ is fed into the Channel Selector module, which discards less important feature channels, resulting in the generation of $F_2$ with fewer but more significant channels. Feeding $F_2$ into the Baseline Network, which is composed of the Feature Extractor module and Regressor module, the $Qs$ is got. The whole framework is trained through a loss function in an end-to-end way. The following sections will provide more details of these modules.

### B. Pre-processing Module

Similar to the IQA methods, which consider image characteristics during framework design, we explore feature characteristics deeply. Given that features are inherently three-dimensional data, we explore both the spatial and channel characteristics. Specifically, due to data correlation, which facilitates the elimination of redundant information while preserving the most important data, we utilize covariance to measure the spatial and channel correlations of features. For clarity, we visualize this process in Fig. 7.

The spatial correlation: Summing up the three-dimensional feature $F \in \mathbb{R}^{H\times W\times C}$ along the channel dimension to obtain two-dimensional data $F' \in \mathbb{R}^{H\times W}$. Subsequently, the covariance matrix $C(F')$ of $F'$ is calculated:

$$C(F') = (c_{ij})_{n\times n} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ c_{n1} & c_{n2} & \dots & c_{nn} \end{pmatrix}, \quad (2)$$

$$\mathrm{c}_{ij} = \mathrm{Cov}\left(F'_i, F'_j\right), i, j = 1, 2, \dots, n, \quad (3)$$

where $F'_i$ and $F'_j$ represent $i$-th row and $j$-th row in $F'$. $c_{ij}$ represents an element of $C(F')$. Cov represents the covariance operation. The variable $n$ in Eqn. (2) and Eqn. (3) equals 14.

The visualization of $C(F')$ is shown in Fig. 9. The pixel value $c_{ij}$ represents the degree of correlation between the $i$-th row and the $j$-th row in $F'$. A higher pixel value corresponds to a redder block, indicating stronger correlation. It is evident that the correlation between rows in $F'$ exists differences.

The channel correlation: Unfolding the three-dimensional feature $F \in \mathbb{R}^{H\times W\times C}$ along the channel dimension and converting each two-dimensional channel into a row vector, $F$ is transformed into two-dimensional data $F'' \in \mathbb{R}^{H'\times W'}$, where $H'$ and $W'$ represent the height and width of $F''$ respectively. The covariance matrix $C(F'')$ of $F''$ is then computed:

$$C(F'') = \left(c'_{ij}\right)_{m\times m} = \begin{pmatrix} c'_{11} & c'_{12} & \dots & c'_{1m} \\ c'_{21} & c'_{22} & \dots & c'_{2m} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ c'_{m1} & c'_{m2} & \dots & c'_{mm} \end{pmatrix}, \quad (4)$$

$$\mathrm{c}'_{ij} = \mathrm{Cov}\left(F''_i, F''_j\right), i, j = 1, 2, \dots, m, \quad (5)$$

where $F''_i$ and $F''_j$ represent $i$-th row and $j$-th row in $F''$. $c'_{ij}$ represents an element of $C(F'')$. Cov represents the covariance operation. The variable $m$ in Eqn. (4) and Eqn. (5) equals 192.

The value of $c'_{ij}$ represents the degree of correlation between the $i$-th row and the $j$-th row in $F''$, where each row of $F''$ corresponds to a feature channel in $F$. Thus, the value of $c'_{ij}$ represents the correlation between feature channels. Fig. 10 shows the visualization results of $C(F'')$. Due to its large size, the entire $C(F'')$ cannot be displayed clearly, thus we provide the partial enlarged map in Fig. 10 (b). The selected portion is enclosed in the red box in Fig. 10 (a). Similar to Fig. 9, a redder block indicates a stronger correlation, revealing varying degrees of correlation between feature channels.

Based on the aforementioned analysis, features exhibit spatial and channel correlations. Regions with strong correlation imply significant associations with other regions, rendering them more important than regions with weak correlation. We highlight the important regions by assigning higher weights. To be specific, inspired by the CBAM [50], we introduce the attention module to emphasize the important spatial and channel areas of features. The attention module pre-processes the input features before sending them into the next part, constituting the Pre-processing module in LFQA. After pre-processing, significant weights are assigned to the critical parts of the input features.

### C. Channel Selector Module

To further explore the critical factors affecting the feature quality, we calculate the channel residuals between the original features and the compressed features as follows:

$$C^i_{res} = |C^i_{ori} - C^i_{com}|, i = 1, 2, ..., 192, \quad (6)$$

where $C^i_{res}$ denotes the $i$-th channel of the feature residual, $C^i_{ori}$ denotes the $i$-th channel of the original feature, and $C^i_{com}$

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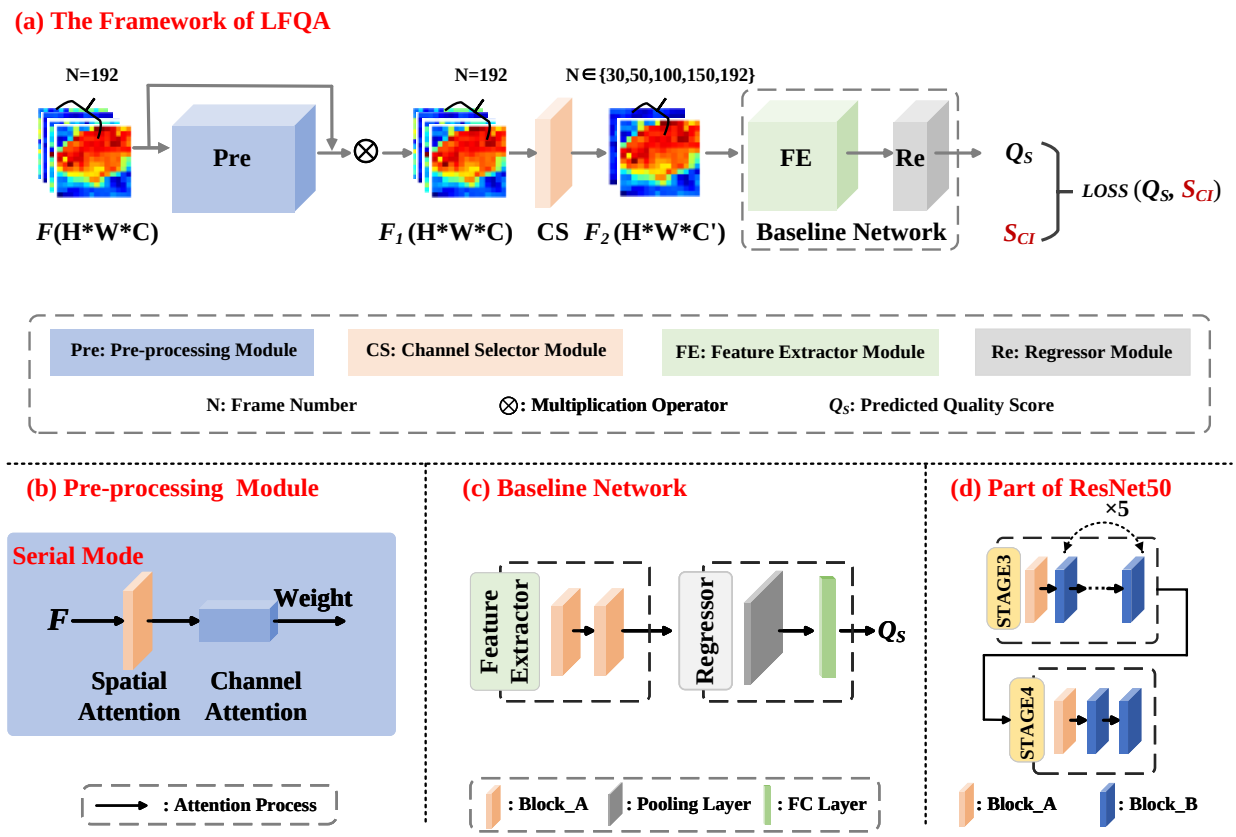


Fig. 8. Illustration of LFQA. LFQA is composed of the Pre module, CS module and Baseline Network. The Pre module weights the $F$ according to the attention mechanism, whose structure is shown in (b). Then, the CS module is connected after the Pre module, which can further remove the channel redundancy of the feature. The CS module is composed of a convolution Layer. The Baseline Network is connected after the CS module and produces the predicted quality score. The specific structure of it is shown in (c). There are two alternative modes in the Pre module: serial mode and parallel mode. Note that we ultimately utilize the serial mode in LFQA, cause it has better performance than parallel mode.

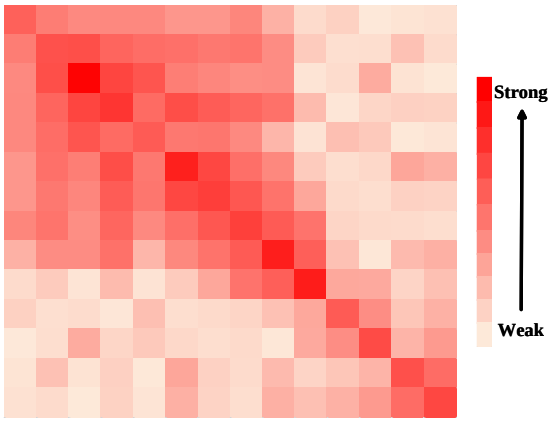


Fig. 9. The visualization result of the $C(F')$. The redder of each square, the higher the corresponding value and the stronger the correlation.

denotes the $i$-th channel of the compressed feature. For ease of visualization, we take the absolute value of the residuals.

The visualization result of $C_{res}$ is depicted in Fig. 11. Each small square represents the residual map of a feature channel, where the pixel value in blue regions is small while the pixel value in red regions is large. It is observable that a few residual channels have more large pixel values, indicating that significant differences exist between the original features and the compressed features across corresponding channels. These differences greatly affect the performance of the coding system. Based on the analysis, only part of the channels play a key role for LFQA. Channels with low importance may contain redundant or even interfering information for subsequent tasks. Thus, feeding all channels to the Baseline Network may not be the optimal strategy.

The output shape after the Pre-processing module remains consistent with the input, thereby keeping the number of feature channels unchanged. This implies that all channels, including those channels with low importance, are sent to the next stage. To selectively transmit only channels with high importance, we introduce the Channel Selector module following the Pre-processing module, which effectively chooses specific channels while automatically discarding those with low importance in an end-to-end manner.

### D. Baseline Network

The existing deep-learning-based IQA methods typically include a feature extractor used to capture essential information that signifies image quality and a regressor that maps this

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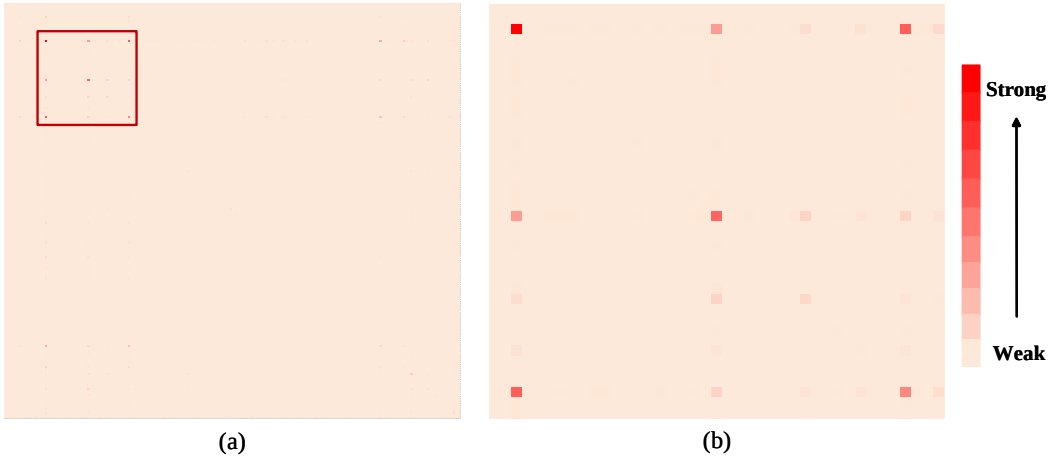


Fig. 10. The visualization results of $C(F'')$. (a) The visualization result of $C(F'')$; (b) The partial enlarged map corresponding to the region circled in the red box in (a). The redder the color, the larger the value, and the stronger the correlation.

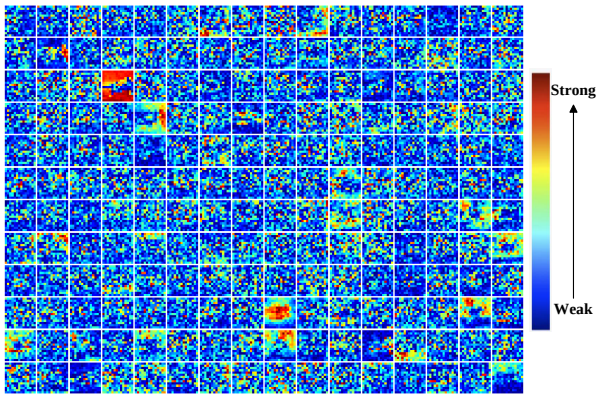


Fig. 11. The visualization result of $C_{res}$. Each square represents one feature residual channel. The redder the color, the greater the difference between the original features and the compressed features across corresponding channels. A few feature channels exhibit significant differences.

information to a quality score. Analogously, deep-learning-based FQA methods also comprise these two important parts. The feature extractor captures the key information that distinguishes the feature quality, while the regressor maps the key information into a quality score.

We choose ResNet50, the most widely used network, as the feature extractor for our proposed method. However, instead of employing the entire ResNet50 network, we only adopt its partial network. Firstly, we aim the FQA method to be lightweight with fast inference speeds, enhancing its practicality. Secondly, since the advanced deep semantic features are more task-specific than shallow features and may not be suitable for quality assessment task. Therefore, we are inclined to extract shallow, more generalized features.

In light of these considerations, we employ the partial network of ResNet50 as the feature extractor for developing a lightweight network for FQA. Specifically, ResNet50 mainly consists of five stages, each stage is composed of Block_A and Block_B, which are two types of residual blocks. Fig. 8 (d) illustrates the structures of STAGE3 and STAGE4 in ResNet50. We select the first blocks from STAGE3 and STAGE4 to form the Feature Extractor module. The Regressor module includes a pooling layer followed by a fully connected (FC) layer, from which the quality score is derived. Fig. 8 (c) shows the specific network architecture of the Baseline Network.

## V. Experiments

In this section, we conduct sufficient experiments to verify the effectiveness of LFQA. In Sec. V-A, we introduce the experimental implementation details. In Sec. V-B, we verify the effectiveness of each module of LFQA. In Sec. V-C, we present the experimental results of IQA methods on the feature database. In Sec. V-D, we demonstrate the lightweight of LFQA. In Sec. V-E, we validate the performance of LFQA on coding system optimization.

### A. Training Setup

We train the proposed method on our feature database. The feature database comprises 39000 features corresponding to 1000 classes of images and differs from conventional image databases used for image quality assessment. Specifically, image databases serve for human vision, focusing on the degree and types of distortion affecting human visual perception, while our feature database serves both human and machine vision and contains two key factors, one is the degree and types of distortions that affect human visual perception, the other is the diversity of object categories utilized to evaluate the classification accuracy. Consequently, the feature database incorporates more factors compared to traditional image databases. To improve the robustness of the proposed model, we adjust the partition ratio, differing from the typical 8:2 ratio for training and testing sets. Approximately 90% of features are allocated for training and the remaining features are used for testing. We set the batch size to 8 and employ the Adam optimizer with an

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initial learning rate of $10^{-5}$. The learning rate follows a "cos" schedule during the 150 training epochs. The experiments are conducted on the NVIDIA GeForce RTX 3080 Ti with 12 GB RAM.

The proposed network undergoes end-to-end training using the loss function shown as follows:

$$L = L_1\left(S_{pre}, S_{GT}\right). \tag{7}$$

We use $L_1$ loss to measure the difference between the predicted score $S_{pre}$ and Ground-Truth $S_{GT}$, where $S_{GT}$ is obtained by Eqn. (1) and equals the $S_{CI}$.

### B. Ablation Studies

Following the established IQA works, we use Spearman Rank-Order Correlation Coefficient (SROCC), Pearson Linear Correlation Coefficient (PLCC), Kendall Rank-Order Correlation Coefficient (KROCC) and Root Mean Square Error (RMSE) as evaluation metrics to assess the correlation between predicted scores and Ground-Truths. SROCC, PLCC and KROCC range from -1 to 1, where negative values indicate negative correlation and positive values signify positive correlation. A larger absolute value indicates a stronger correlation and superior performance. RMSE assesses the difference between predicted scores and Ground-Truths, with smaller values indicating better performance.

*1) Results from the Baseline Network:* We feed the test features into the well-trained Baseline Network to obtain the predicted scores. Calculating the PLCC, SROCC, KROCC and RMSE between the predicted scores and Ground-Truths, the experimental results are obtained and shown in Table. I. Both PLCC and SROCC are around 0.7. To satisfy the demand for more accurate assessment, further performance enhancement is necessary.

*2) Contribution of the Pre-processing Module:* Due to the three-dimensional features exhibiting both spatial correction and channel correction, the Pre-processing module consists of spatial and channel attention modules. Different combinations of these modules may impact the performance of the proposed method. Hence, we carefully consider these combinations and categorize them into two types: serial and parallel. In parallel mode, the input $F$ is simultaneously processed by the Channel Attention module and the Spatial Attention module, and the Weight is determined by combining the outputs of the two attention modules. In serial mode, the input $F$ first undergoes the Spatial Attention module and then is fed into the Channel Attention module. The Weight is obtained after the Channel Attention module.

Table. I presents the experimental results of the Pre-processing module in both serial and parallel modes. It is obvious that the Pre-processing module significantly enhances performance. Compared to the Baseline Network, the PLCC, SROCC, and KROCC are all improved by an average of 5% - 6%. The performance enhancement proves the effectiveness of the Pre-processing module. Besides, there are differences in the experimental results between the serial and parallel modes. Specifically, while the parallel mode outperforms the serial mode, the advantage is not substantial, indicating that attention module combinations have a relatively minor impact on the proposed network.

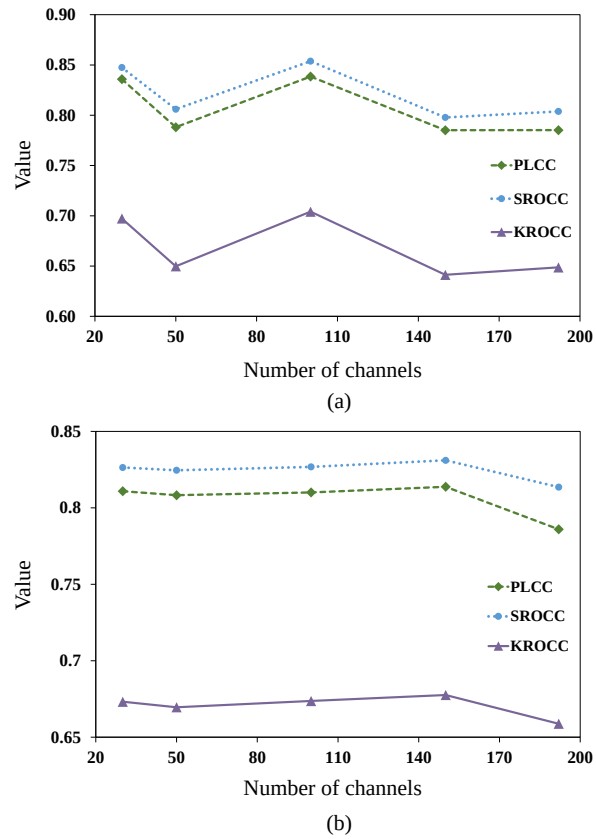


Fig. 12. The visualization of the experimental results of the Channel Selector Module. (a) The results under serial mode; (b) The results under parallel mode. The horizontal axis denotes the number of channels, while the vertical axis represents the PLCC, SROCC, and KROCC values.

*3) Contribution of the Channel Selector Module:* To further mitigate information redundancy within features, we select partial feature channels after the Pre-processing module for further processing. To verify the effectiveness of the Channel Selector module, we conduct sufficient ablation experiments varying the number of selected channels, including 150, 100, 50, and 30, noting that the original channel number is 192. The experimental results are shown in Table. I. Compared to the Baseline Network, the PLCC, SROCC, and KROCC are all improved by an average of 7% - 8%. The improvement of performance proves the effectiveness of the Channel Selector module.

*4) Experimental Results of Combining All Modules Under Different Modes:* We sequentially combine all modules and validate the performance of the proposed method under different modes. The Pre-processing module, as the upstream module, we need to carefully consider its impact. Thus, we provide experimental results for different channel numbers under serial and parallel modes respectively. The experimental results are shown in Table. II and are visualized in Fig. 12. Furthermore, to verify the effectiveness of the Channel Attention module and Spatial Attention module individually, we provide experimental results across different channel numbers

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TABLE I
QUANTITATIVE RESULTS OF EVALUATING THE CONTRIBUTION OF PRE AND CS IN LFQA. BN DENOTES THE BASELINE NETWORK, PRE(SM) DENOTES THE PRE-PROCESSING MODULE IN SERIAL MODE, PRE(PM) REPRESENTS THE PRE-PROCESSING MODULE IN PARALLEL MODE, AND CS (·) REPRESENTS THE CS MODULE UNDER DIFFERENT CHANNEL NUMBERS. ↑ MEANS THAT THE BIGGER THE BETTER. ↓ MEANS THAT THE SMALLER THE BETTER.

| No. | BN | Pre | CS | Mode | PLCC ↑ | SROCC ↑ | KROCC ↑ | RMSE ↓ |
|---|---|---|---|---|---|---|---|---|
| 1 | ✓ | | | BN | 0.7275 | 0.7486 | 0.5935 | 1.4772 |
| 2 | ✓ | ✓ | | BN+Pre(SM) | 0.7852 | 0.8038 | 0.6487 | 0.8997 |
| 3 | ✓ | ✓ | | BN+Pre(PM) | 0.7860 | 0.8136 | 0.6588 | 0.9321 |
| 4 | ✓ | | ✓ | BN+CS(30) | 0.8088 | 0.8242 | 0.6701 | 0.8176 |
| 5 | ✓ | | ✓ | BN+CS(50) | 0.8103 | 0.8237 | 0.6678 | 0.8134 |
| 6 | ✓ | | ✓ | BN+CS(100) | 0.8188 | 0.8340 | 0.6812 | 0.7963 |
| 7 | ✓ | | ✓ | BN+CS(150) | 0.7983 | 0.8174 | 0.6629 | 0.8371 |

TABLE II
EXPERIMENTAL RESULTS OF COMBINING ALL MODULES UNDER DIFFERENT MODES AND DIFFERENT CHANNEL NUMBERS. ↑ MEANS THAT THE BIGGER THE BETTER. ↓ MEANS THAT THE SMALLER THE BETTER. THE TOP RESULTS ARE HIGHLIGHTED IN **BOLD**.

| Mode | Channel Number | PLCC ↑ | SROCC ↑ | KROCC ↑ | RMSE ↓ |
|---|---|---|---|---|---|
| Spatial Attention | 30 | 0.2790 | 0.2800 | 0.2052 | 1.8543 |
| | 50 | 0.3309 | 0.3418 | 0.2537 | 2.3426 |
| | 100 | 0.4836 | 0.5039 | 0.3746 | 2.9963 |
| | 150 | 0.4859 | 0.5137 | 0.3849 | 2.4546 |
| | 192 | 0.8361 | 0.8528 | 0.7019 | 0.7610 |
| Channel Attention | 30 | 0.8265 | 0.8374 | 0.6841 | 0.8129 |
| | 50 | 0.8303 | 0.8439 | 0.6930 | 0.7733 |
| | 100 | 0.8302 | 0.8442 | 0.6920 | 0.7732 |
| | 150 | 0.7841 | 0.7989 | 0.6430 | 0.8674 |
| | 192 | 0.8181 | 0.8386 | 0.6846 | 0.7991 |
| Parallel Combination | 30 | 0.8109 | 0.8264 | 0.6732 | 0.8129 |
| | 50 | 0.8083 | 0.8246 | 0.6696 | 0.8184 |
| | 100 | 0.8101 | 0.8268 | 0.6737 | 0.8135 |
| | 150 | 0.8138 | 0.8311 | 0.6776 | 0.8069 |
| | 192 | 0.7860 | 0.8136 | 0.6588 | 0.9005 |
| Serial Combination | 30 | 0.8359 | 0.8475 | 0.6973 | 0.7611 |
| | 50 | 0.7881 | 0.8060 | 0.6498 | 0.8846 |
| | 100 | **0.8385** | **0.8538** | **0.7041** | **0.7556** |
| | 150 | 0.7851 | 0.7978 | 0.6413 | 0.8643 |
| | 192 | 0.7852 | 0.8038 | 0.6487 | 0.8997 |

for each module, which are included in Table. II as well. Several conclusions are drawn from the experimental results:

- From Table. I, the contributions of the Pre-processing module and the Channel Selector module are approximately comparable. However, combining both modules yields the best experimental results.
- The serial and parallel modes affect the experimental results. The trends of PLCC, SROCC, and KROCC are different in the two modes. Notably, the optimal performance is achieved with 100 channels in serial mode, yielding an SROCC of 0.8538.
- The experimental results indicate that the Channel Attention module outperforms and demonstrates greater consistency compared to the Spatial Attention module across different channel numbers, suggesting that Channel Attention module plays a more important role in LFQA. These experimental results can be attributed to the inherent characteristics of the features. As depicted in Fig. 10, only a minority of channels exhibit strong correlations and serve a critical function. When reducing the number of channels, the Channel Attention module effectively preserves the important channels, ensuring consistent performance.

### C. Comparison of IQA Methods on the Feature Database

To verify the performance of IQA methods on our feature database, we conduct the following experiments. Due to substantial differences between the shape and characteristics of features and those of images, directly feeding features into IQA methods is both unfeasible and unreasonable. Thus,

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TABLE III
EXPERIMENTAL RESULTS OF COMPARING METHODS AND OUR METHOD ON THE FEATURE DATABASE. ↑ MEANS THAT THE BIGGER THE BETTER. ↓ MEANS THAT THE SMALLER THE BETTER. SPEED REPRESENTS THE INFERENCE TIME FOR AN IMAGE. THE TOP TWO RESULTS ARE HIGHLIGHTED IN **BOLD** AND UNDERLINE.

| Methods | PLCC ↑ | SROCC ↑ | KROCC ↑ | RMSE ↓ | Param (M)↓ | Speed (s)↓ | FLOPs (M)↓ |
|---|---|---|---|---|---|---|---|
| CNNIQA [51] | 0.7680 | 0.7873 | 0.6316 | 1.0098 | **1.1930** | 0.0167 | 22356 |
| HyperIQA [52] | 0.8115 | 0.8226 | 0.6685 | 0.8177 | 27.9680 | 0.0124 | 11770 |
| DEIQT [53] | 0.6715 | 0.7058 | 0.5598 | 1.0942 | 22.7740 | 0.0042 | 4256 |
| VRCNet [54] | 0.5533 | 0.5785 | 0.4469 | 1.2300 | 11.4130 | 0.3747 | 10271 |
| MNIQA [55] | 0.3744 | 0.3813 | 0.2825 | 3.2432 | 127.7370 | 0.0270 | 108616 |
| CDF [7] | - | - | - | - | 24.1070 | 0.0077 | 12131 |
| Pro | **0.8385** | **0.8538** | **0.7041** | **0.7556** | 7.5960 | **0.0040** | **178.6** |

we first choose deep-learning-based IQA methods rather than traditional IQA methods, mitigating the discrepancies between features and images through retraining. Then, we rescale features to the same resolution as the images before feeding them into the IQA methods. In addition, we modify the channel number in the first layer of the IQA methods to accommodate the features into the selected IQA frameworks.

We select five classical and high-performance deep-learning-based IQA methods, including CNNIQA [51], hyperIQA [52], DEIQT [53], VCRNet [54] and MANIQA [55], and retrain them on our feature database. Similar to Sec. V-B, we also use SROCC, PLCC, KROCC and RMSE as evaluation metrics to assess the correlation between predicted scores and Ground-Truths. The experimental results are shown in Table. III.

Although the IQA methods achieve reasonably good performance after retraining, they still underperform compared to the proposed method because they do not account for the characteristics of the features. Our method shows improvements over the best-performing IQA method (hyperIQA) in the SROCC, PLCC, KROCC, and RMSE, with a notable 3% increase in SROCC, demonstrating the effectiveness of the proposed method.

### D. Comparison of Run-time Complexity

*1) Comparison with CDF:* Actually, the performance of human and machine tasks can be obtained by executing the CDF framework directly. Considering this, we focus on the lightweight when designing the FQA method, so that the performance of the two tasks can be got faster with fewer resources. The input of LFQA consists of the compressed (reconstructed) features. For a more fair comparison, we select only a part of the CDF, which is referred to as the *Decoder* in Fig. 4, and remove the remaining part. This ensures that the input of the CDF is also comprised of the compressed (reconstructed) features, and the experimental results of CDF are obtained on the selected part. We measure the model parameter, inference time and FLOPs (floating-point operations per second) of the two methods using the same hardware, and report them in Table. III. The proposed LFQA has fewer parameters and FLOPs, and is nearly *2x* faster than the CDF,

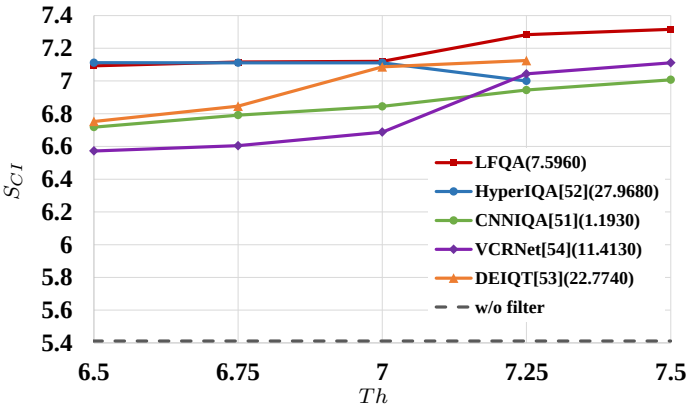


Fig. 13. Coding system performance of LFQA compared with IQA methods under different $Th$ on FDB. w/o filter represents the results without filtering, reflecting the original performance of the coding system. The number in parentheses represents the model parameter.

indicating that LFQA is better suited for resource-constrained environments.

*2) Comparison with IQA methods:* We also compare our LFQA with the selected IQA methods in terms of inference time, model parameter, and FLOPs using the same hardware, and the experimental results are also shown in Table. III. Our LFQA outperforms all selected IQA methods in terms of both inference time and FLOPs. As for the model parameter, our LFQA ranks second among all comparison methods. Overall, our LFQA is more lightweight than the comparison methods while achieving optimal performance, demonstrating the effectiveness and superiority of it.

### E. Performance on Coding System Optimization

As shown in Fig. 3, we use LFQA to replace the LAM for filtering out those features that do not meet the requirements, thereby enhancing the coding system performance. Specifically, we set multiple performance thresholds $Th_s$ based on the Ground-Truths of our feature database, which range from 1 to 8. Considering the actual application scenarios, $Th_s$ that are too low do not effectively filter features and are not meaningful, while higher $Th_s$ are more relevant. Therefore, we set the lower limit of the $Th_s$ to 6.5 and the upper limit

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to 7.5, employing uniform sampling in between. The final set of $Th_s$ are 6.5, 6.75, 7, 7.25, and 7.5.

We compare LFQA with the IQA methods discussed in Sec. V-C, and the experimental results are presented in Fig. 13. The results using the "filtering" mechanism outperform those without it, demonstrating the effectiveness of employing quality assessment methods as the filter. Furthermore, LFQA exhibits superior performance compared to the IQA methods, highlighting its advantages. It is worth noting that there are no results for MNIQA, since its predicted results are clustered around a specific value in our feature database, making it unsuitable for being a filter. Additionally, the predicted results of HyperIQA and DEIQT are all below 7.5. Therefore, there are no data points for them in Fig. 13 when the $Th$ is set to 7.5. These experimental results illustrate the potential of LFQA for optimizing coding system performance and combined with its lightweight nature, underscoring the feasibility and necessity of LFQA.

## VI. Conclusion

In this paper, we conduct a preliminary exploration of feature quality assessment and demonstrate its feasibility and necessity. Specifically, we build a feature database, propose an indicator for assessing the feature quality, and present a lightweight feature quality assessment (LFQA) method. To improve the accuracy of LFQA, we thoroughly explore the inherent characteristics of the features during framework design. Sufficient experimental results prove the effectiveness of the proposed method. However, the proposed LFQA represents an initial step, with unresolved challenges remaining in the field of feature quality assessment. The indicator for features is essential, as it provides a benchmark and guideline for optimizing FQA methods. As a preliminary attempt, we combine each indicator for human and machine tasks in a simple yet effective manner. In the future, we intend to explore more combinations tailored to practical application scenarios. Besides, more generic feature quality assessment methods need to be explored to meet more human and machine tasks, further optimizing and improving the coding system.

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**Shipei Wang** received the M.E. degree from Ningbo University, Zhejiang, China, in 2019. She is currently pursuing a PhD at the School of Communication and Information Engineering, Shanghai University. Her current research interests include video coding for machine, feature quality assessment , and image quality assessment.

**Ping An** (Member, IEEE) received the B.E. and M.E. degrees from the Hefei University of Technology, Hefei, China, in 1990 and 1993, respectively, and the Ph.D. degree from Shanghai University, Shanghai, China, in 2002. Since 1993, she has been affiliated with Shanghai University, where she currently holds the position of Professor in the Video Processing Group, School of Communication and Information Engineering. From 2011 to 2012, she was a Visiting Researcher with the Communication Systems Group, Technische University at Berlin, Germany. She has published more than 200 refereed technical papers in international journals and conferences. Her research interests include image and video processing, with a focus on immersive intelligent video processing. She was a recipient of the Second Prize of the Shanghai Municipal Science and Technology Progress Award, the Second Prize in Natural Sciences of the Ministry of Education, and the Second Prize in Natural Sciences of the Chinese Institute of Electronics.

**Chao Yang** received the B.E. and Ph.D. degrees from the School of Communication and Information Engineering, Shanghai University, Shanghai, China, in 2012 and 2017, respectively. From Nov. 2017 to Oct. 2018, he was a Post-Doctoral Fellow with the School of Electrical Engineering System, University of Southern California, Los Angeles, CA, USA. He is currently an Associate Professor with the School of Communication and Information Engineering, Shanghai University. His current research interests include video processing, video compression, and image quality assessment.

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**Gongyang Li** received the Ph.D. degree from Shanghai University, Shanghai, China, in 2022. He is currently an Associate Professor at the School of Communication and Information Engineering, Shanghai University, Shanghai, China. From 2021 to 2022, he was a Visiting Ph.D. Student with the School of Computer Science and Engineering, Nanyang Technological University, Singapore. From 2022 to 2024, he worked as a Postdoctor at Shanghai University, Shanghai, China. His research interests include saliency detection, multi-model image processing, and image/video segmentation.

**Xinpeng Huang** (Member, IEEE) received B.S. and M.E. degrees from Zhengzhou University of Light Industry in 2013 and 2016 and a Ph.D. degree from Shanghai University in 2019. He started his postdoctoral career at Shanghai University as a candidate for the 2020 Shanghai Super Postdoctoral Incentive Program. He is currently an Associate Professor with Shanghai University, Shanghai, China. His research interests include image and video processing, with a focus on light field coding, enhancement, assessment.

**Shiqi Wang** (Senior Member, IEEE) received the Ph.D. degree in computer application technology from Peking University in 2014. He is currently an Associate Professor with the Department of Computer Science, City University of Hong Kong. He has proposed more than 70 technical proposals to ISO/MPEG, ITU-T, and AVS standards. He authored or coauthored more than 300 refereed journal articles/conference papers, including more than 100 IEEE Transactions. His research interests include semantic and visual communication; AI-generated content management; machine learning; information forensics and security; and image/video quality assessment. He received the Best Paper Award from IEEE VCIP 2019, ICME 2019, IEEE Multimedia 2018, and PCM 2017. His coauthored article received the Best Student Paper Award in the IEEE ICIP 2018. He served or serves as an Associate Editor for IEEE TCSVT, TMM, TIP, TCyber, IEEE Access, and APSIPA Transactions on Signal and Information Processing.

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